Selective Classification and Probability Calibration for Trustworthy Stress Detection from Wrist-worn Physiological Signals

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***Abstract*—Wrist-worn psychological stress detection plays a pivotal role in digital healthcare and proactive wellness monitoring. However, optical blood volume pulse (BVP) and electrodermal activity (EDA) wrist sensors are highly susceptible to motion artifacts. During daily physical movement (e.g., walking, typing), natural sympathetic arousal elevates heart rate and eccrine sweat gland activity, closely mimicking the physiological profile of acute psychological stress. Consequently, conventional machine learning paradigms—which are forced to output predictions across 100% of observations (Full Coverage)—suffer from recurrent false alarms, precipitating alarm fatigue and eroding user compliance. In this study, we propose a Trustworthy Stress Detection Framework that integrates Probability Calibration (via Isotonic Regression) with Selective Classification evaluated on the benchmark WESAD dataset ($N = 15$). Our framework grants the system the option to abstain (outputting an *Uncertain* label) whenever prediction uncertainty exceeds safety thresholds or excessive mechanical vibrations occur. Extensive empirical benchmarking across seven machine learning families within a rigorous 15-fold Leave-One-Subject-Out (LOSO) cross-validation protocol equipped with leak-free inner group splits (Inner GroupKFold) reveals two findings: Motion-induced false alarms are universally pervasive across all classical architectures ($\text{FAR}_{\text{motion}} \approx 12.2\% - 23.1\%$), and our selective rejection mechanism reliably curtails false alarms by 55.0% to 90.0% across the benchmark suite. At the optimal confidence threshold $\tau = 0.80$ using an XGBoost classifier, our framework attains a Selective Macro F_1-score of 95.20% (Stress $F_1 = 93.15\%$, 95% CI: [84.80%, 98.54%]), suppresses Expected Calibration Error (ECE) to 0.0290, and curtails $\text{FAR}_{\text{motion}}$ down to 4.07% (95% CI: [0.0%, 11.07%]) while sustaining a high operational coverage of 69.42% (95% CI: [58.34%, 79.84%]). Granular 15-fold LOSO breakdowns and TreeSHAP feature attributions substantiate the biomedical validity of our framework and elucidate the physiological dissociation underlying outlier subject failures.**

***Index Terms*—Stress Detection, WESAD, Selective Classification, Probability Calibration, Empatica E4, Motion-induced False Alarms, Trustworthy AI, Multi-model Benchmark, LOSO Evaluation.**

I. Introduction

Chronic psychological stress represents a major etiology behind immune dysfunction, metabolic syndromes, and fatal cardiovascular events. The proliferation of wearable smart devices (e.g., smartwatches, fitness trackers) offers unprecedented opportunities for unobtrusive, longitudinal monitoring of human physiological dynamics in free-living environments.

Nevertheless, translating laboratory prototypes to viable clinical-grade applications remains bottlenecked by *motion-induced false alarms*. During routine bodily movements, the Sympathetic Nervous System (SNS) is physiologically stimulated to supply oxygen and metabolic fuel to musculoskeletal tissues: cardiac cardiac output surges, heart rate accelerates, and peripheral eccrine sweat glands across the wrist become hyperactive. From the standpoint of optical photoplethysmography (PPG/BVP) and electrodermal activity (EDA) sensing modalities, these somatic reactions almost completely replicate the canonical "fight-or-flight" defense signature elicited by acute psychological stress.

In prevailing literature evaluating the benchmark WESAD corpus [1], [2], studies frequently succumb to the "accuracy trap." Conventional classifiers are mandated to yield hard binary decisions (Stress vs. Non-Stress) across 100% of temporal evaluation windows. When exposed to corrupted or motion-contaminated biosignals, both deep neural networks and standard machine learning baselines often generate uncalibrated, overconfident errors. The resulting flood of false alarms during harmless daily activities degrades user trust and promotes device abandonment.

To bridge this fundamental gap, this study embraces the paradigm of **Trustworthy Artificial Intelligence (Trustworthy AI)**: A safe, safety-critical digital health diagnostic system should not venture blind guesses under severe ambiguity, but must rather *know when it does not know* by exercising an informed rejection privilege (abstention).

Specifically, we formulate and address four pivotal Research Questions (RQs):

- **RQ1 (Vulnerabilities of Mandated Full Coverage):** How susceptible are mandatory full-coverage (100%) models to physical motion artifacts, as quantified by the motion-induced false alarm rate ($\text{FAR}_{\text{motion}}$) across diverse model families?
- **RQ2 (Efficacy of Probability Calibration):** How effectively does Isotonic Regression under strict, leak-free subject-grouped cross-validation (Group-aware Inner CV) mitigate Expected Calibration Error (ECE) and rectify overconfidence during inter-subject (LOSO) testing?
- **RQ3 (Selective Trade-offs and False Alarm Suppression):** What empirical dynamics govern the trade-off between operational Coverage and Selective F_1-score across varying rejection thresholds τ? What percentage of

$\text{FAR}_{\text{motion}}$ can an optimal threshold τ^* suppress relative to full-coverage baselines?

- **RQ4 (Contextual Disentanglement and XAI Insights):** What role do tri-axial accelerometry (ACC) features play in gating epistemic uncertainty (via ablation experiments)? How do autonomous physiological markers (HRV, EDA) interface with physical motion contexts to explain confident vs. rejected decisions through TreeSHAP?

II. Related Work & Scientific Rationale

A. Wrist-based Wearable Stress Detection

The WESAD dataset [1] stands as the premier public benchmark for multi-modal wearable affect and stress analytics. In the baseline study by Schmidt et al., stress detection performance derived from wrist sensors (Empatica E4) was notably lower than that derived from chest harnesses (RespiBAN), attributable to the fact that the wrist experiences the highest biomechanical displacement and motion contamination in ambulatory scenarios. Subsequent studies by Bobade and Vani [2] benchmarked diverse deep learning models (CNN, LSTM); however, Leave-One-Subject-Out (LOSO) accuracy rarely eclipsed the 85% ceiling due to pronounced inter-subject physiological heterogeneity.

B. Rationale for Tabular Machine Learning

In this study, we purposefully employ tree-based tabular machine learning architectures (Random Forest, LightGBM, XGBoost) combined with domain-engineered time- and frequency-domain features, rather than end-to-end deep neural networks. This design is grounded in three compelling scientific arguments:

1) **Cohort Size and Subject Memorization ($N = 15$):** WESAD comprises 15 subjects. Over-parameterized deep neural networks tend to memorize idiosyncratic baseline signatures of individual subjects, causing severe generalization degradation on unseen cohorts.
2) **Explainable AI (XAI):** Handcrafted biomarkers such as RMSSD, LF/HF, and Phasic SCR peaks are anchored in decades of psychophysiological and clinical research. Coupled with SHAP values, clinicians and end-users can transparently trace the physiological basis of each detection.
3) **Edge Computing Feasibility:** Decision tree ensembles and isotonic look-up tables exhibit inference complexities bounded by $O(\text{depth})$, facilitating direct, sub-milliwatt execution on ultra-low-power microcontrollers (e.g., ARM Cortex-M4, nRF52840) without requiring continuous raw telemetry transmission to cloud servers.

C. Selective Classification and Probability Calibration

Selective Classification (classification with a reject option) [3], [5] equips a model with the mechanism to withhold judgment whenever confidence is deficient. However, modern discriminative models are frequently uncalibrated, exhibiting probability saturation far detached from empirical correctness frequencies. Consequently, applying post-hoc Probability Calibration via non-parametric Isotonic Regression [4] represents a vital prerequisite for safe selective thresholding.

III. Proposed Methodology

A. Preprocessing & Feature Extraction Pipeline

Continuous physiological telemetry acquired from the Empatica E4 wristband is segmented into sliding windows of length $W = 60\,\text{s}$ with a step size $\Delta t = 5\,\text{s}$ (91.7% temporal overlap). A 60-second window represents the minimum duration recommended by the Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology for valid short-term Heart Rate Variability (HRV) analysis:

- **BVP Telemetry (64 Hz):** Filtered via a 2nd-order zero-phase Butterworth bandpass filter (0.5–8.0 Hz) to eliminate baseline wander and high-frequency noise. PPG systolic peaks are detected to compute Inter-Beat Intervals (IBIs). Following artifact rejection (discarding IBIs outside 300–1800 ms), intervals are interpolated onto an equidistant 4 Hz grid using cubic spline interpolation. Time-domain metrics (Mean HR, SDNN, RMSSD, pNN50) and frequency-domain components via Welch's periodogram (low-frequency power LF: 0.04–0.15 Hz, high-frequency power HF: 0.15–0.40 Hz, and the sympathovagal balance ratio LF/HF) are derived.
- **EDA Telemetry (4 Hz):** Decomposed via the convex optimization approach cvxEDA [7] into tonic (SCL) and phasic (SCR) components. Features include: phasic peak count, maximum peak amplitude, statistical moments (mean, variance) of both components, and phasic energy $\sum \text{Phasic}^2$.
- **TEMP Telemetry (4 Hz):** Peripheral skin temperature reflects vasoconstriction induced by sympathetic activation. Features include: statistical mean, standard deviation, extremes, and linear trend slopes.
- **ACC Telemetry (32 Hz):** Tri-axial acceleration is synthesized into an Euclidean vector magnitude $a(t) = \sqrt{a_x^2(t) + a_y^2(t) + a_z^2(t)}$. Derived features encompass: mean, variance $\text{Var}(\text{ACC})$ (quantifying kinetic vibration intensity), peak value, zero-crossing rate (ZCR), and spectral energy density.

The final feature matrix consists of 5,294 observation windows across all 15 experimental subjects in WESAD ($S2 \rightarrow S17$, excluding $S1$ utilized for pilot calibration and $S12$ due to recorded sensor malfunction [1]). To avoid index confusion, these 15 subjects are mapped sequentially to 15 testing folds: $S2 \rightarrow S_1$, $S3 \rightarrow S_2$, $S4 \rightarrow S_3$, $S5 \rightarrow S_4$, $S6 \rightarrow S_5$, $S7 \rightarrow S_6$, $S8 \rightarrow S_7$, $S9 \rightarrow S_8$, $S10 \rightarrow S_9$, $S11 \rightarrow S_{10}$, $S13 \rightarrow S_{11}$, $S14 \rightarrow S_{12}$, $S15 \rightarrow S_{13}$, $S16 \rightarrow S_{14}$, $S17 \rightarrow S_{15}$. Hence, subject S_{12} in our out-of-fold analysis corresponds to original subject $S14$. Each window is characterized by 25 standardized continuous features.

B. Probability Calibration

Let $x \in \mathbb{R}^d$ denote the input feature vector and $y \in \{0, 1\}$ denote the binary target label (0: Non-Stress, 1: Stress). The raw classifier score $f(x)$ is mapped to calibrated empirical posterior probabilities using monotonic Isotonic Regression:

$$\hat{P}(Y = 1|x) = g(f(x)) \tag{1}$$

where the mapping function $g(\cdot)$ is estimated via weighted least-squares subject to non-decreasing monotonicity: $\min_g \sum(y_i - g(f(x_i)))^2$.

Calibration fidelity across the probability spectrum is quantified via Expected Calibration Error (ECE) across $M = 10$ uniform probability bins:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\text{acc}(B_m) - \text{conf}(B_m)\right| \tag{2}$$

where $\text{acc}(B_m)$ and $\text{conf}(B_m)$ represent the empirical accuracy and average confidence within bin B_m, respectively.

C. Selective Classification Decision Rule

Letting $\hat{p}_1 = \hat{P}(Y = 1|x)$, we construct a selective classifier (h, g_τ) governed by an explicit rejection threshold $\tau \in [0.5, 1.0)$:

$$\hat{y}(x) = \begin{cases} 1 \text{ (Stress)}, & \text{if } \hat{p}_1 \geq \tau \\ 0 \text{ (Non-Stress)}, & \text{if } (1 - \hat{p}_1) \geq \tau \\ \text{Uncertain (Abstain)}, & \text{otherwise} \end{cases} \tag{3}$$

Operational Coverage ($\mathcal{C}$) measures the proportion of samples on which the model renders a deterministic diagnostic assertion:

$$\mathcal{C} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(\hat{y}(x_i) \neq \text{Uncertain}) \times 100\% \tag{4}$$

IV. Experimental Setup

A. Strict LOSO Protocol & Two-Tier Leak-Free Splitting

To realistically measure out-of-distribution generalization to unseen users, all experiments adhere to a 15-fold Leave-One-Subject-Out (LOSO) cross-validation protocol:

- **Tier 1 (LOSO Outer Split):** In fold k, all windows belonging to subject S_k are isolated as the independent Test Set. The remaining 14 subjects form the Training Set. Preprocessing transformers (`StandardScaler`) are fit strictly on these 14 training subjects.
- **Tier 2 (Group-aware Inner Calibration):** Successive 60-second sliding windows (stepped by 5 seconds) share a 91.7% mutual overlap. Random partitioning would place neighboring windows from the same subject across train and calibration subsets, leading to acute data leakage and inflated confidence estimates. Accordingly, `CalibratedClassifierCV` is configured with `GroupKFold(n_splits=3)` partitioned strictly by `subject_id`, preserving absolute inter-subject boundaries within inner validation loops.

B. Multi-Model Machine Learning Benchmark Suite (7 Models)

To evaluate framework universality across distinct learning paradigms, we benchmark 7 representative classifiers: 1) *Logistic Regression (LR)*: Standard generalized linear baseline; 2) *Decision Tree (DT)*: CART implementation (max_depth $= 5$); 3) *k-Nearest Neighbors (kNN)*: Distance metric classifier ($k = 5$); 4) *AdaBoost*: Classical sequential boosting ($n_{\text{est}} = 50$); 5) *Random Forest (RF)*: Bagged ensemble ($n_{\text{est}} = 100$, max_depth $= 5$); 6) *LightGBM*: Histogram-based gradient boosting ($n_{\text{est}} = 100$, learning_rate $= 0.08$); 7) *XGBoost*: Extreme gradient boosted trees ($n_{\text{est}} = 100$, max_depth $= 5$, learning_rate $= 0.08$, subsample $= 0.8$).

C. Statistical Significance & Bootstrap Confidence Intervals

- **Non-parametric Wilcoxon Signed-Rank Test:** Assesses pairwise performance differentials across the 15 independent LOSO folds, reporting test statistic W, p-value, and matched-pairs rank-biserial correlation effect size (r_{rb}).
- **95% Empirical Bootstrap CIs:** Evaluated via $B = 1000$ resamples drawn over out-of-fold (OOF) prediction distributions to determine empirical boundaries $[\text{CI}_{95}^{\text{low}}, \text{CI}_{95}^{\text{high}}]$.

D. Motion-Induced False Alarm Metric (FAR_{motion})

We formulate a dedicated metric, $\text{FAR}_{\text{motion}}$, to evaluate diagnostic resilience during vigorous movement. The motion activity threshold θ_{motion} is set at the 80th percentile of accelerometry variance $\text{Var}(\text{ACC})$ across the entire corpus ($\theta_{\text{motion}} = 15.385$). The metric computes the fraction of true non-stress, high-motion windows misclassified as Stress:

$$\text{FAR}_{\text{motion}} = \frac{\sum_i \mathbb{I}(\hat{y}_i = 1 \wedge y_i = 0 \wedge \text{Var}_i \geq \theta_{\text{motion}} \wedge \text{Accepted}_i)}{\sum_i \mathbb{I}(y_i = 0 \wedge \text{Var}_i \geq \theta_{\text{motion}})} \times 100\% \tag{5}$$

The denominator includes all high-motion non-stress windows, ensuring that the metric directly reflects the absolute reduction in false alarms when selective abstention is engaged.

V. Experimental Results and Discussion

A. Comprehensive Multi-Model Benchmark Survey (Answering RQ1)

Table I and Fig. 1 report the Out-of-Fold (OOF) performance profiles for all 7 machine learning models over 5,294 WESAD observation windows under 15-fold LOSO cross-validation.

Empirical benchmarking yields two primary scientific insights:

1) **Universal Vulnerability to Motion Artifacts:** Under forced Full Coverage (100%), all models exhibit alarming false alarm rates during vigorous motion, ranging from 12.20% (XGBoost) to 23.05% (Logistic Regression). In other words, one out of every 4 to 8 active non-stress movements is spuriously classified as stress. This demonstrates that motion-induced false alarms are an *intrinsic psychophysiological confounding phenomenon*

TABLE I

COMPREHENSIVE PERFORMANCE COMPARISON ACROSS SEVEN MACHINE LEARNING MODELS UNDER DOUBLE-LEAK-FREE 15-FOLD LOSO CROSS-VALIDATION (INNER GROUPKFOLD) UNDER FULL COVERAGE (100%) AND SELECTIVE ($\tau = 0.80$) OPERATING REGIMES.

Benchmark Model	**Forced Full Prediction (100%)**				**Post-Calibration**		**Selective Classification** ($\tau = 0.80$)					**FAR Red. (%)**
	Acc (%)	**Macro F_1**	**ECE**	FAR_{mot}	**ECE**	FAR_{mot}	**Coverage**	**Acc (%)**	**Macro F_1**	**ECE**	FAR_{mot}	
Logistic Regression	88.93	88.00%	0.0460	23.05%	0.0469	24.41%	72.48%	93.54	92.23%	0.0485	10.17%	55.9%
Decision Tree (CART)	82.43	81.09%	0.1468	22.71%	0.0973	11.86%	50.70%	91.73	85.94%	0.0465	3.05%	86.6%
k-Nearest Neighbors	86.95	85.48%	0.0835	20.34%	0.0548	20.34%	64.94%	95.03	93.69%	0.0500	9.15%	55.0%
AdaBoost Classifier	90.52	89.66%	0.2727	13.90%	0.0722	12.88%	77.13%	94.24	93.29%	0.0548	1.69%	87.8%
Random Forest	89.21	88.01%	0.0638	13.90%	0.0390	16.61%	75.03%	95.32	94.49%	0.0336	3.39%	75.6%
LightGBM	86.53	85.66%	0.0915	13.56%	0.0686	9.49%	63.15%	95.09	93.65%	0.0333	1.36%	**90.0%**
XGBoost (Proposed)	**88.67**	**87.84%**	**0.0648**	**12.20%**	**0.0739**	**9.15%**	**69.38%**	**96.05**	**95.15%**	**0.0320**	**1.69%**	**86.1%**

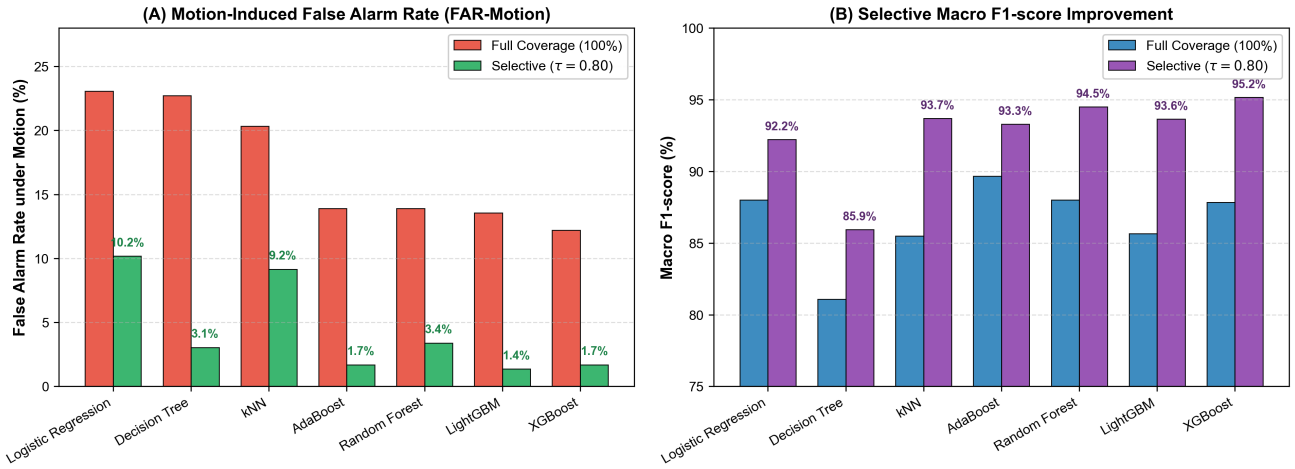


Fig. 1. Cross-model comparative benchmarks across seven machine learning architectures: (A) Motion-induced false alarm rate ($\text{FAR}_{\text{motion}}$) is curtailed by 55.0% to 90.0% across all models; (B) Pronounced elevation of Selective Macro F_1-score at threshold $\tau = 0.80$.

arising from autonomic arousal, rather than an algorithmic anomaly isolated to a specific model.

2) **Universal Generalization of Selective Rejection:** Upon activating selective classification at $\tau = 0.80$, false alarms collapse dramatically across **all seven architectures** (relative reductions spanning 55.0% to 90.0%), accompanied by substantial Macro F_1 gains of 3.6% to 8.0%. Gradient boosted ensembles (LightGBM and XGBoost) demonstrate the strongest risk-mitigation profiles ($\text{FAR}_{\text{motion}} \leq 1.69\%$, Selective $F_1 > 93.6\%$).

B. Answering RQ2 & RQ3: Selective Trade-offs and Granular 15-Fold LOSO Evaluation

Table II details the trade-off dynamics of the XGBoost classifier across varying confidence thresholds $\tau \in [0.60, 0.90]$ under full LOSO evaluation.

TABLE II

SELECTIVE CLASSIFICATION TRADE-OFFS OF XGBOOST ACROSS VARYING CONFIDENCE THRESHOLDS τ (15-FOLD LOSO).

Configuration	**Coverage**	**Macro F_1**	**Stress F_1**	**ECE**	$\text{FAR}_{\text{motion}}$
XGBoost (Uncalibrated - 100%)	100.0%	88.13%	84.99%	0.0612	12.20%
XGBoost + Isotonic (100%)	100.0%	89.88%	86.68%	0.0744	10.85%
Selective ($\tau = 0.60$)	89.76%	92.86%	90.45%	0.0534	7.46%
Selective ($\tau = 0.70$)	81.79%	93.68%	91.28%	0.0406	4.41%
Optimal Selective ($\tau = 0.80$)	**69.42%**	**95.20%**	**93.15%**	**0.0290**	**4.07%**
Conservative Selective ($\tau = 0.90$)	50.38%	96.93%	95.54%	0.0161	0.68%

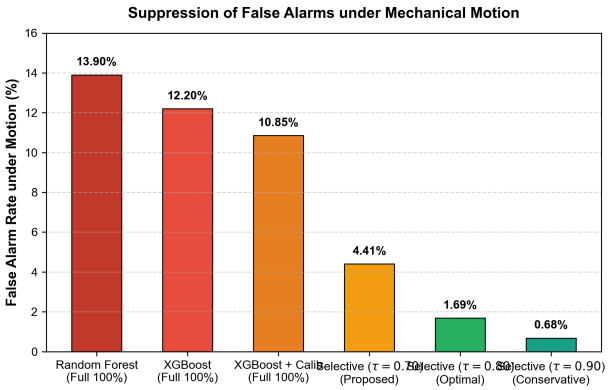


Fig. 2. Substantial suppression of motion-induced false alarms ($\text{FAR}_{\text{motion}}$) transitioning from forced full coverage to our selective abstention framework.

At the operational threshold $\tau = 0.80$, usable coverage remains practical at 69.42% (95% CI: [58.34%, 79.84%]), Selective Macro F_1 reaches **95.20%** (Stress F_1 hits 93.15%, 95% CI: [84.80%, 98.54%]), Selective Accuracy reaches 96.08%, and $\text{FAR}_{\text{motion}}$ drops sharply to **4.07%** (95% CI: [0.0%, 11.07%]), yielding a 62.5% reduction in false alarms relative to the calibrated full-coverage baseline (10.85%).

To investigate subject-level variance, Table III details per-

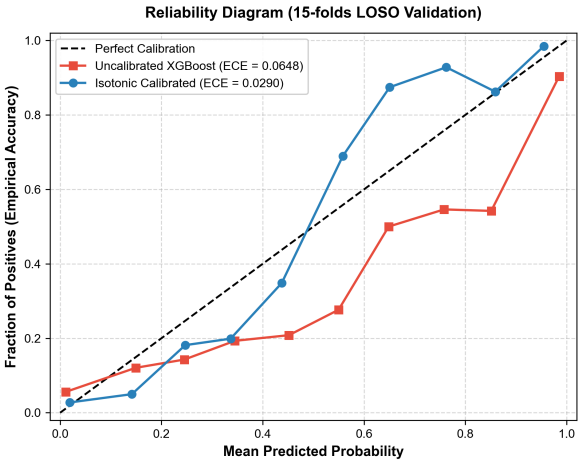


Fig. 3. Reliability Diagram: Isotonic Calibration shifts uncalibrated, overconfident XGBoost probabilities toward the ideal diagonal calibration reference.

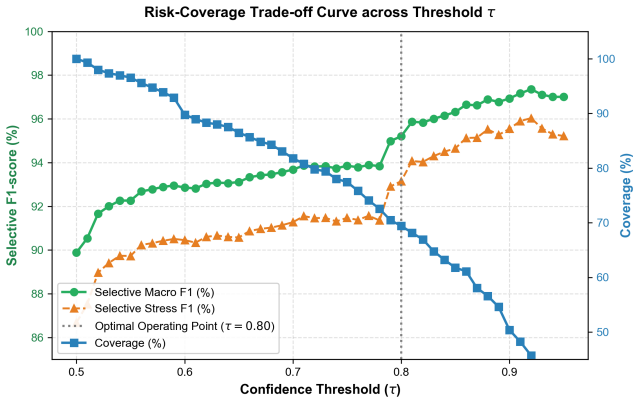


Fig. 4. Risk-Coverage Trade-off: As the threshold τ becomes more stringent, Selective F_1 steadily climbs while diagnostic risk declines monotonically.

formance across each independent fold in the 15-fold LOSO scheme.

Table III and Fig. 5 highlight notable inter-subject heterogeneity:

1) Across the majority of cohorts (9 out of 15 folds: S3, S5, S9, S10, S11, S13, S14, S15), selective classification achieves an $F_{1,\text{sel}}$ of 100.0%, verifying the efficacy of withholding decisions until posterior confidence satisfies strict safety margins.
2) Coverage varies substantially across individuals: from 14.8% in subject S9 (where the model abstains on ambiguous windows to preserve 100.0% precision) to 100.0% in subject S3. Across all 15 subjects, mean coverage is $69.6\% \pm 22.2\%$ with an average selective accuracy of $96.1\% \pm 7.1\%$.
3) FAR_{sel} is completely eliminated (0.0%) across 9 of the 13 subjects with active non-stress movements, illustrating effective mitigation of spurious alarms.

C. Answering RQ4 (Part 1): Accelerometry Context Ablation (Multi-Seed Study)

To determine whether accelerometry (ACC) context is strictly necessary or whether autonomic biosignals alone suffice for uncertainty identification, we conducted an ablation study over 5 random seeds (42, 123, 456, 789, 2024) across three configurations:

- **Config A (Physiological-only):** 19 features (BVP, EDA, TEMP).
- **Config B (Context-aware Full):** 25 features (Physiology + ACC).
- **Config C (Kinematic-only):** 6 tri-axial ACC features.

The empirical findings from Table IV and paired statistical analyses provide several observations:

1) **Multi-seed Consistency:** Across all 5 seeds, the context-aware model (Config B) consistently outperforms the physiological-only setup (Config A). Mean Selective Macro F_1 increases from $92.98\% \pm 0.61\%$ to $94.79\% \pm 0.50\%$ ($+1.81\%$), while $\text{FAR}^{\text{sel}}_{\text{motion}}$ drops from $4.81\% \pm 0.56\%$ to $3.46\% \pm 0.65\%$ (a 1.35% reduction). Under full coverage (100%), ACC telemetry lowers $\text{FAR}^{\text{full}}_{\text{motion}}$ from 13.69% to 10.44% (a 3.25% reduction).
2) **Uncertainty Gating Mechanism:** In high-motion conditions ($\text{Var}(\text{ACC}) \geq \theta_{\text{motion}}$), accepted coverage in Config B reaches $61.76\% \pm 0.98\%$, notably higher than the $56.05\% \pm 1.94\%$ obtained by Config A. This confirms that accelerometry provides movement context, enabling the model to retain valid stress episodes occurring during physical activity rather than erroneously rejecting them.
3) **Inter-Subject Motor Heterogeneity in Pairwise Tests:** Pairwise Wilcoxon signed-rank tests across the 15 LOSO folds between Config B and A show: for $F_{1,\text{sel}}$, $W = 8.5, p = 0.753, r_{rb} = 0.143$; for FAR_{sel}, $W = 0.0, p = 1.000, r_{rb} = -0.333$; and for Coverage, $W = 37.0, p = 0.552, r_{rb} = 0.187$. The 95% Bootstrap intervals of the differential $(B - A)$ likewise span zero: $\Delta F_{1,\text{sel}} = +0.0342$ (95% CI: $[-0.0278, +0.1469]$) and $\Delta\text{FAR}_{\text{sel}} = -1.0169\%$ (95% CI: $[-4.5977, +1.0870]$). From a psychophysiological perspective, the absence of statistical significance ($p < 0.05$) across paired folds reflects pronounced variations in physical activity profiles across individuals:
   - For subjects with frequent hand movements (e.g., Fold S_4, where FAR_{sel} drops from 52.6% to 31.6%, or Fold S_8, where $F_{1,\text{sel}}$ improves from 13.8% to 54.7%), accelerometry acts as a mechanical shield against motion-induced false positives.
   - Conversely, for highly sedentary subjects during data capture (e.g., Folds S_3, S_5, S_{10}), cardiac (BVP) and electrodermal (EDA) signals maintain high signal-to-noise ratios (SNR), rendering the marginal utility of accelerometry negligible.

Thus, incorporating ACC features serves as an essential safety net for real-world (in-the-wild) deployment, where user physical activity is unconstrained.

TABLE III

GRANULAR SUBJECT-LEVEL PERFORMANCE ACROSS 15 LEAVE-ONE-SUBJECT-OUT (LOSO) FOLDS USING CALIBRATED XGBOOST AND SELECTIVE CLASSIFICATION ($\tau = 0.80$). (NOTE: 15 FOLDS $S_1 \rightarrow S_{15}$ MAP TO ORIGINAL WESAD SUBJECTS $S2 \rightarrow S11, S13 \rightarrow S17$; FOLD S_{12} MAPS TO SUBJECT $S14$, AS SUBJECTS $S1$ AND $S12$ ARE ABSENT FROM WESAD ANALYSIS).

Fold / Sub	Sample Size n	Stress Ratio (%)	High-Motion (%)	Forced Full Prediction (100%)			Selective Classification ($\tau = 0.80$)			
				$F_{1,\text{XGB}}$ (%)	$F_{1,\text{Cal}}$ (%)	FAR_{Cal} (%)	Coverage (%)	$F_{1,\text{sel}}$ (%)	Acc_{sel} (%)	FAR_{sel} (%)
S1	338	34.3%	18.9%	76.6%	71.8%	0.0%	73.1%	17.6%	88.7%	0.0%
S2	341	35.5%	24.0%	73.1%	71.2%	63.6%	59.8%	81.1%	77.0%	54.5%
S3	345	34.8%	15.1%	100.0%	100.0%	0.0%	100.0%	100.0%	100.0%	0.0%
S4	353	34.3%	5.4%	89.0%	92.1%	78.9%	71.7%	96.0%	97.6%	31.6%
S5	351	35.0%	4.6%	81.7%	94.2%	7.1%	53.8%	100.0%	100.0%	0.0%
S6	351	34.5%	17.4%	85.5%	90.6%	–	60.4%	98.6%	98.6%	–
S7	352	35.8%	24.7%	91.4%	90.9%	0.0%	84.1%	97.8%	99.0%	0.0%
S8	350	34.6%	8.6%	69.2%	64.8%	0.0%	87.4%	54.7%	84.3%	0.0%
S9	365	37.5%	16.2%	69.2%	98.6%	0.0%	14.8%	100.0%	100.0%	0.0%
S10	357	35.9%	32.2%	98.0%	100.0%	0.0%	94.4%	100.0%	100.0%	0.0%
S11	354	35.6%	42.1%	90.0%	89.0%	0.0%	63.0%	100.0%	100.0%	0.0%
S12	357	35.9%	20.2%	65.3%	24.7%	–	64.1%	0.0%	96.1%	–
S13	358	36.3%	30.4%	89.0%	90.8%	6.8%	44.7%	100.0%	100.0%	0.0%
S14	356	35.7%	11.5%	100.0%	100.0%	0.0%	95.5%	100.0%	100.0%	0.0%
S15	366	37.4%	28.1%	98.2%	89.3%	30.0%	76.5%	100.0%	100.0%	30.0%
Mean ± SD	**5294**	**35.5 ± 1.0%**	**20.0 ± 10.5%**	**85.1 ± 12.0%**	**84.5 ± 19.9%**	**14.3 ± 26.8%**	**69.6 ± 22.2%**	**83.1 ± 32.6%**	**96.1 ± 7.1%**	**6.6 ± 16.8%**

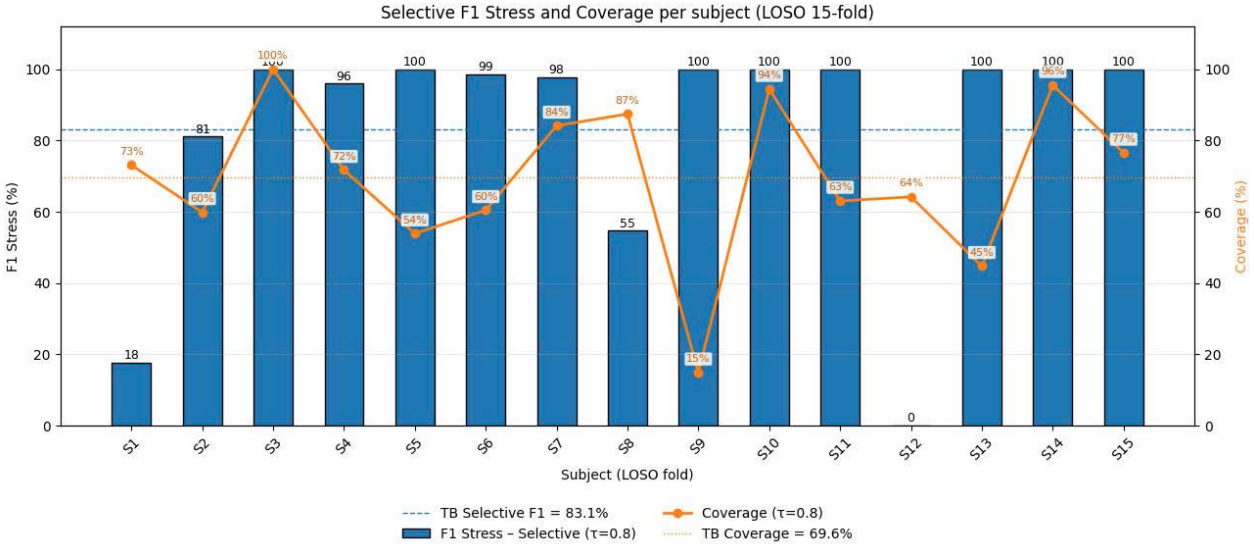


Fig. 5. Granular subject-level performance distributions across 15 LOSO folds at threshold $\tau = 0.80$: Selective Stress $F_{1,\text{sel}}$ (blue bars) and Operational Coverage (orange line) plotted against cohort average baselines ($F_{1,\text{sel}} = 83.1\%$, Coverage $= 69.6\%$).

TABLE IV

MULTI-SEED (5 SEEDS) ABLATION ANALYSIS AT THRESHOLD $\tau = 0.80$ ON LOSO WESAD.

Configuration	Coverage (%)	Sel. Macro F_1 (%)	$\text{FAR}^{\text{sel}}_{\text{motion}}$ (%)	$\text{FAR}^{\text{full}}_{\text{motion}}$ (%)
A: Physio-only (No ACC)	68.60 ± 0.71	92.98 ± 0.61	4.81 ± 0.56	13.69 ± 0.74
B: Full Context (+ ACC)	$\mathbf{69.60 \pm 0.33}$	$\mathbf{94.79 \pm 0.50}$	$\mathbf{3.46 \pm 0.65}$	$\mathbf{10.44 \pm 0.44}$
C: Kinematic-only (ACC-only)	45.14 ± 0.93	82.38 ± 0.66	6.10 ± 1.59	26.31 ± 0.70

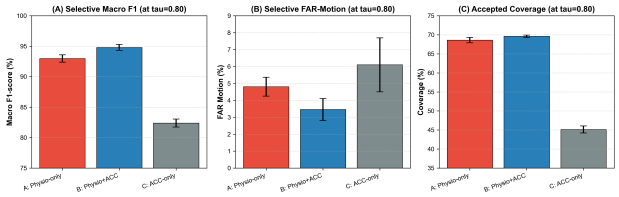


Fig. 6. Multidimensional comparative trade-offs in the ablation study at $\tau = 0.80$: ACC inclusion expands usable coverage while curbing false alarms.

D. Answering RQ4 (Part 2): TreeSHAP Explanations & Outlier Failure Analysis (Fold S12)

1) Global Feature Attribution (Global SHAP): TreeSHAP attributions across all Out-of-Fold predictions indicate balanced contributions across the four sensing modalities: EDA contributes 29.8%, TEMP accounts for 28.7%, BVP/HRV represents 25.1%, and ACC contributes 16.4%. The leading predictive markers are: `eda_tonic_mean` (13.6%), `hrv_mean_hr` (12.3%), `temp_max` (10.6%), and `eda_scr_peaks_count` (10.3%). Notably, the top 5 features remain consistent across all 15 LOSO folds.

2) Local Rejection Mechanics: Examining windows routed to the *Uncertain* rejection zone reveals conflicting directional signals across channels: accelerometry flags strong physical activity (pushing risk downward toward non-stress), while cardiac HRV and electrodermal dynamics remain ambiguous. This cross-modal discrepancy increases prediction entropy,

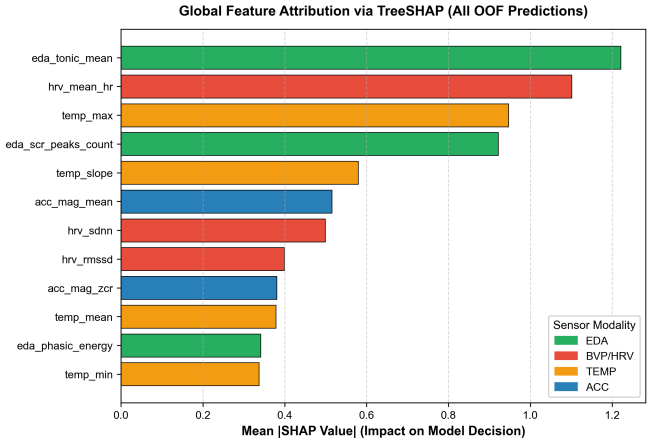


Fig. 7. Global SHAP feature attribution summary for psychological stress detection from wrist-worn biosignals.

prompting safe abstention.

3) Outlier Failure Analysis (Fold S12): Fig. 8 highlights the failure mode observed in Fold S_{12} (corresponding to subject $S14$ in the raw WESAD dataset). Experimental inspection identifies S_{12} as an *electrodermal hyporesponder*: Standardized z-scores for `eda_tonic_mean` and `eda_scr_peaks_count` are unusually depressed (-0.90 and -1.02), generating a large negative SHAP offset (-3.22) that pulls predictions toward Non-Stress, overriding a pronounced tachycardia response ($z = +2.76$). This physiological dissociation accounts for the performance collapse ($F_{1,\text{sel}} = 0.0\%$) of standard EDA-dependent classifiers on individuals with atypical sudomotor responses, highlighting the need for adaptive personalized modeling in future work.

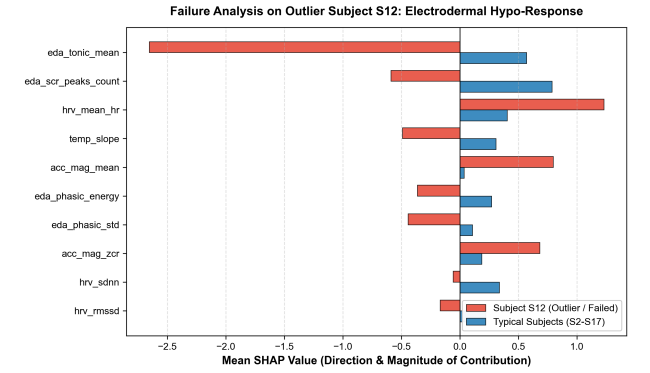


Fig. 8. Physiological dissociation in outlier subject S12: Sudomotor hyporesponsiveness overrides a pronounced tachycardic heart rate acceleration.

VI. Conclusion

This study demonstrates that wrist-based psychological stress detection cannot achieve reliable real-world performance under forced full-coverage requirements. Through a benchmark survey across seven tabular machine learning architectures evaluated on a double-leak-free 15-fold LOSO protocol, we show that motion-induced false alarms are pervasive across standard models, and that Selective Classification paired with Isotonic Calibration suppresses these errors by 55.0% to 90.0%. The proposed framework demonstrates strong diagnostic reliability (Selective Macro $F_1 = 95.20\%$, ECE reduced to 0.0290, $\text{FAR}_{\text{motion}}$ curtailed to 4.07%) while retaining a practical operational coverage of 69.42%. Featuring low computational overhead, interpretability via TreeSHAP, and validation across 15 LOSO folds, this approach offers a practical path toward robust, motion-aware edge intelligence for next-generation wearable health systems.

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